Copied vs Timestomped Files

Digital Forensics & Incident Response

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Executive Summary

This project demonstrates how NTFS stores timestamp metadata and how forensic analysts can distinguish between normal file activity, copied files, and deliberate timestamp manipulation (timestomping).

Using a controlled Windows 11 laboratory, files were created, copied, and backdated before acquiring the $MFT with FTK Imager. Analysis in MFT Explorer v2026.5.0 focused on comparing $STANDARD_INFORMATION (SI) and $FILE_NAME (FN) timestamps alongside forensic indicators such as Copied and uSecZeros to identify anti-forensic activity.

Case Information

Property	Value
Case ID	DFIR-2026-NTFS-014
Investigator	Julian Derry
Evidence Type	NTFS $MFT
Operating System	Windows 11
Filesystem	NTFS
Acquisition Tool	FTK Imager
Analysis Tool	MFT Explorer v2026.5.0

Objectives

- Examine NTFS timestamp behavior during file creation.
- Identify timestamp changes after a file is copied.
- Demonstrate how timestomping alters SI timestamps.
- Compare SI and FN metadata to detect manipulation.
- Validate forensic indicators within the $MFT.

Scenario

A suspected insider attempts to conceal activity by copying sensitive documents and backdating a file to make it appear older than it actually is.

The objective is to determine whether NTFS metadata can reliably distinguish legitimate file operations from deliberate timestamp manipulation.

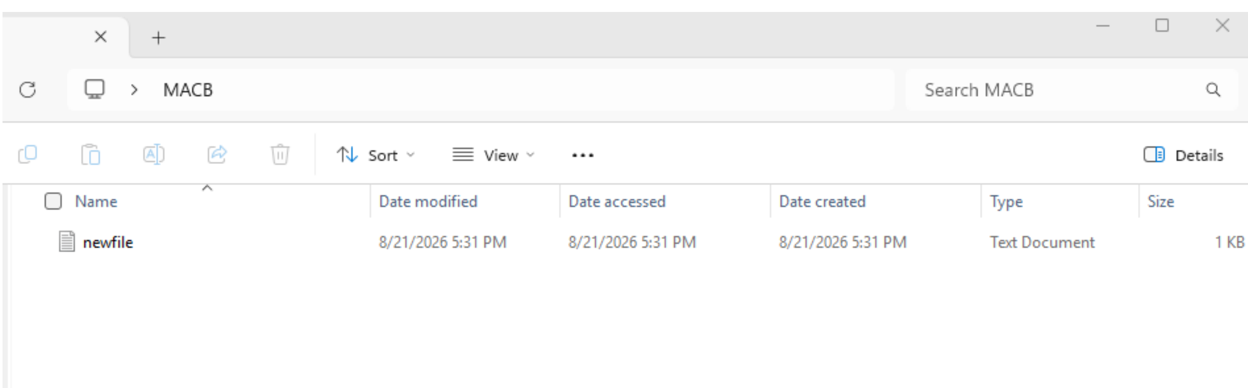
Methodology

Phase 1: Create a New File

A text file was created using Command Prompt. Immediately after creation, all three Windows timestamps were identical because the file had not yet been modified or accessed.

Timestamp	Value
Date Created	2026-08-21 5:31 PM
Date Modified	2026-08-21 5:31 PM
Date Accessed	2026-08-21 5:31 PM

Evidence



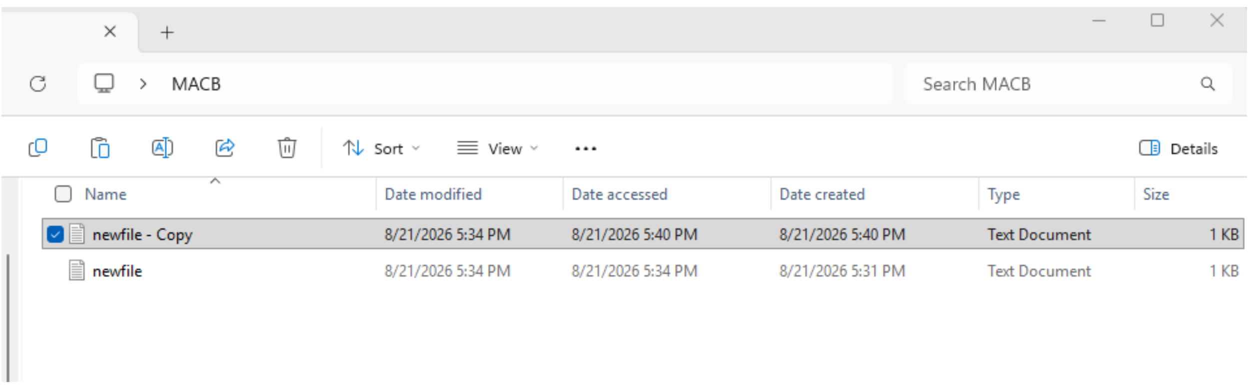
Phase 2: Copy the File

The file was copied to another location. NTFS preserved the Date Modified timestamp from the original file while assigning new Date Created and Date Accessed values to the copied file.

Timestamp	Original File	Copied File
Date Created	Original	Current time
Date Modified	Original	Original
Date Accessed	Original	Current time

This behavior provides a reliable forensic indicator that a file was copied rather than newly created.

Evidence



Phase 3: Perform Timestomping

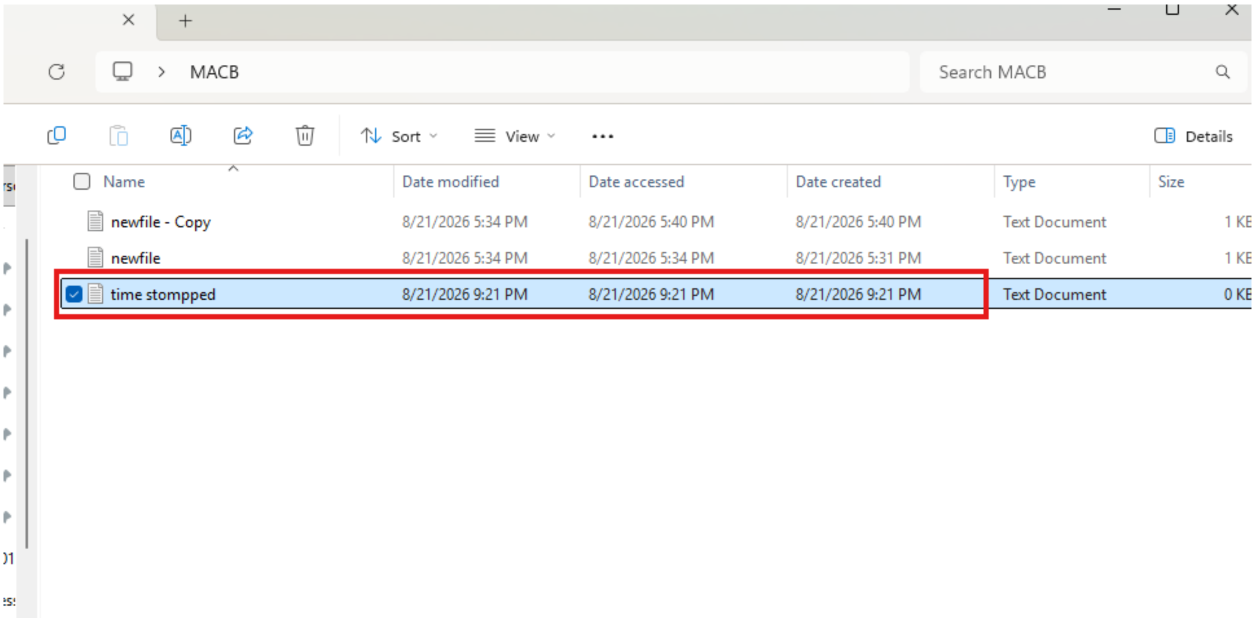
A new file named timestomped.txt was created and its timestamps were intentionally backdated using Command Prompt.

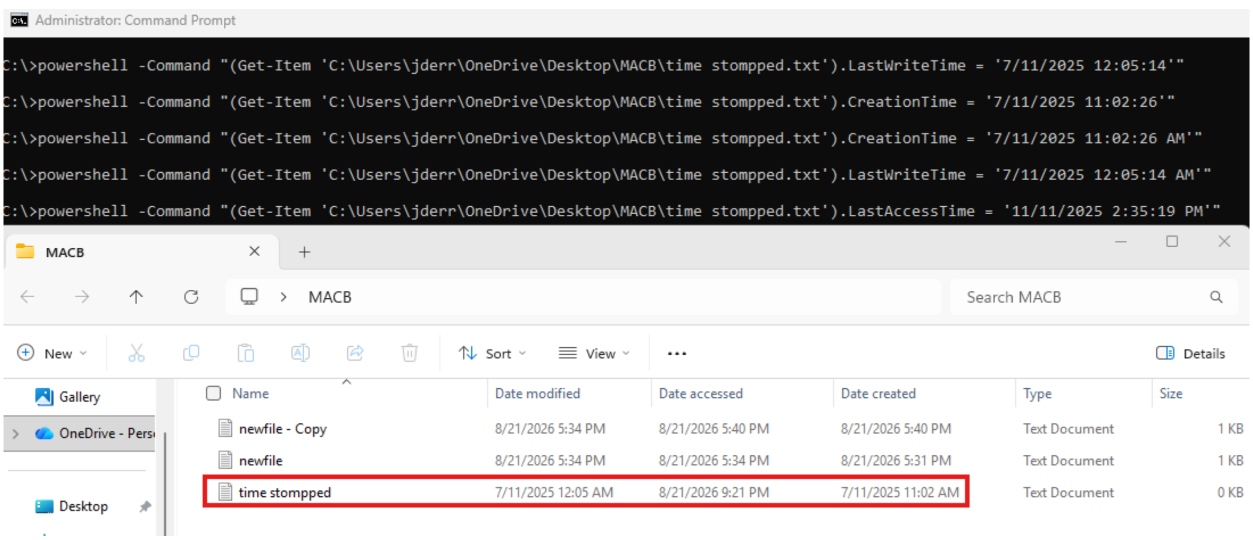
The following timestamps were altered:

- Date Created
- Date Modified
- Date Accessed

This simulates a common anti-forensic technique used to mislead timeline reconstruction during an investigation.

Evidence





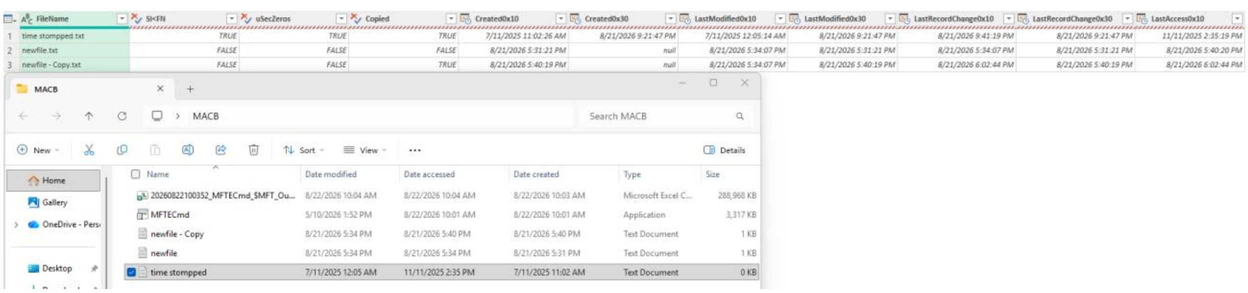
Phase 4: MFT Analysis

The NTFS $MFT was exported using FTK Imager and examined with MFT Explorer. The directory containing the test files was filtered to compare metadata between the copied and timestomped files.

Field	Forensic Significance
Copied	Indicates whether the file record was created through a copy operation
SI<FN	Shows SI timestamps occurring earlier than FN timestamps, a strong timestomping indicator
uSecZeros	Microsecond values set to zero, commonly associated with artificial timestamp modification

By correlating these fields, the manipulated file was clearly distinguished from the legitimately copied file.

Evidence



	FileName	SI<FN	uSecZeros	Copied	Created0x10	Created0x30	LastModified0x10	LastModified0x30	LastRecordChange0x10	LastRecordChange0x30	LastAccess0x10
1	time stompped.txt	TRUE	TRUE	TRUE	7/11/2025 11:02:26 AM	8/21/2026 9:21:47 PM	7/11/2025 12:05:14 AM	8/21/2026 9:21:47 PM	8/21/2026 9:41:19 PM	8/21/2026 9:21:47 PM	11/11/2025 2:35:19 PM
2	newfile.txt	FALSE	FALSE	FALSE	8/21/2026 5:31:21 PM	null	8/21/2026 5:34:07 PM	8/21/2026 5:31:21 PM	8/21/2026 5:34:07 PM	8/21/2026 5:31:21 PM	8/21/2026 5:40:20 PM
3	newfile - Copy.txt	FALSE	FALSE	TRUE	8/21/2026 5:40:19 PM	null	8/21/2026 5:34:07 PM	8/21/2026 5:40:19 PM	8/21/2026 6:02:44 PM	8/21/2026 5:40:19 PM	8/21/2026 6:02:44 PM

Name	Date modified	Date accessed	Date created	Type	Size
20260822100352_MFTECmd_$MFT_Ou...	8/22/2026 10:04 AM	8/22/2026 10:04 AM	8/22/2026 10:03 AM	Microsoft Excel C...	288,968 KB
MFTECmd	5/10/2026 1:52 PM	8/22/2026 10:01 AM	8/22/2026 10:01 AM	Application	3,317 KB
newfile - Copy	8/21/2026 5:34 PM	8/21/2026 5:40 PM	8/21/2026 5:40 PM	Text Document	1 KB
newfile	8/21/2026 5:34 PM	8/21/2026 5:34 PM	8/21/2026 5:31 PM	Text Document	1 KB
time stompped	7/11/2025 12:05 AM	11/11/2025 2:35 PM	7/11/2025 11:02 AM	Text Document	0 KB

Key Findings

Artifact	Observation
Normal File	SI and FN timestamps align naturally
Copied File	Copied = TRUE
Timestomped File	SI<FN = TRUE
Microsecond Analysis	uSecZeros = TRUE
MFT Evidence	Metadata differentiates copied files from manipulated files

DFIR Takeaways

- Copied files do not duplicate every timestamp.
- NTFS preserves Date Modified while generating a new Date Created timestamp.
- Timestomping primarily targets $STANDARD_INFORMATION (SI) attributes.
- Comparing SI and FN timestamps is an effective method for identifying timestamp manipulation.
- $MFT metadata often reveals anti-forensic activity that Windows Explorer properties alone cannot.

Conclusion

This investigation demonstrates that NTFS metadata remains highly valuable even when an attacker attempts to manipulate timestamps. While timestomping can alter the visible $STANDARD_INFORMATION values, inconsistencies between SI and FN attributes, together with indicators such as Copied and uSecZeros, provide strong evidence of file manipulation.

The key lesson is that investigators should rely on $MFT analysis rather than Windows file properties alone. Correlating multiple NTFS metadata attributes produces a far more accurate reconstruction of file history and helps expose common anti-forensic techniques.

Tools Used

- FTK Imager
- MFT Explorer v2026.5.0
- Windows Command Prompt
- Windows 11 (NTFS)